

Module 3 (Connectivity) Project

(Develop any ONE)

1. Project title: AI/ML-Based Predictive Maintenance and Vehicle Health Monitoring System

AI/ML Technologies: Random Forest, XGBoost, LSTM, XAI

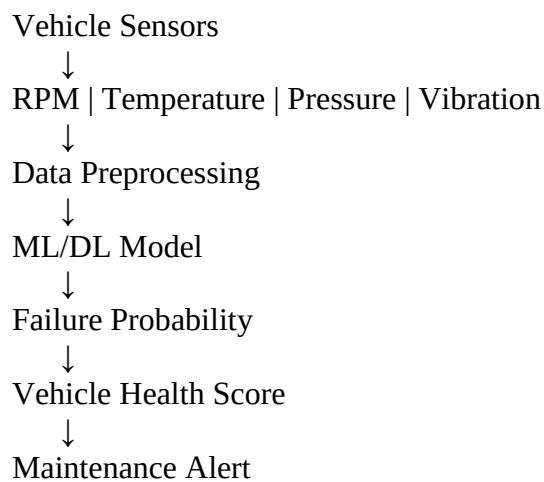
Models: Logistic Regression, Decision Tree, Random Forest, XGBoost, LSTM, GRU, Transformer

Goal: Predict vehicle/component failure before it occurs.

Input data

- Engine temperature
- RPM
- Oil pressure
- Vibration
- Battery voltage
- Coolant temperature
- Fuel consumption
- Vehicle speed
- Operating hours

Architecture



Sample Output

Vehicle Health: 72%
Failure Probability: 68%
Status: WARNING

Major contributing factors:

1. High vibration

2. Rising temperature
3. Abnormal oil pressure

2. Project title: AI-Based Driver Behaviour, Fatigue and Risk Detection System

AI/ML Technologies: ML, CNN, LSTM

The system learns driving behavior and identifies potentially unsafe situations.

Parameters

- Speed
- Acceleration
- Braking
- Steering
- Lane deviation
- Eye closure
- Head position
- Phone usage

Develop a system that identifies:

- Normal driving
- Aggressive acceleration
- Sudden braking
- Distracted driving
- Drowsiness
- Unsafe driving

Architecture

Camera + Vehicle Sensors
↓
Feature Extraction
↓
CNN/LSTM
↓
Driver Behavior
↓
Risk Score

Possible classes

- Normal
- Aggressive
- Distracted
- Drowsy
- Unsafe braking
- Unsafe acceleration

3. Project title: **AI-Based Automotive Review and Customer Sentiment Analytics**

AI/ML Technologies: BERT, XLM-R, NLP

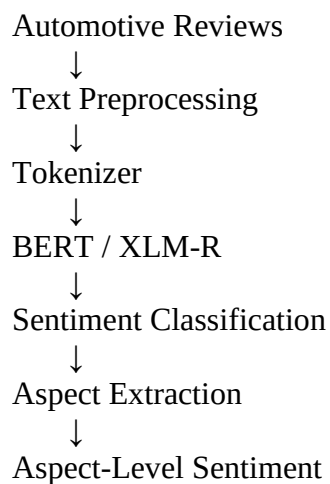
Goals: Analyze vehicle reviews and identify sentiment for individual aspects.

Data

Automotive reviews can be analyzed for:

- Vehicle
- Engine
- Battery
- Mileage
- Safety
- Comfort
- Service
- Infotainment
- Price

Architecture



Example:

"The battery range is excellent, but the charging time is too long."

Output

Aspect	Sentiment
Battery range	Positive
Charging time	Negative